THE NAIVE COTANGENT COMPLEX DOES NOT DEPEND ON THE CHOSEN PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-modules/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The complex attached to a presentation of a sheaf of algebras as a quotient of a polynomial algebra, and the complex defined by the identity presentation. The set-up paragraph is reproduced verbatim as a context span of this card (source lines 5922-5939). Fixes the two objects that the statement compares.
- The compatibility of the formation of differentials with taking stalks, and the description of the naive cotangent complex of a presentation over a ring. Invoked inside the counted proof at source lines 5966-5972 as pointers to a lemma of this chapter and to a section of another chapter. Reduces the required quasi-isomorphism to the corresponding statement in commutative algebra.

Source reference. Original lemma lemma-abelian: sources/stacks-modules/pinned-original-source.tex, source lines 439–439. Navigation pointer only: the counted text refers to the Stacks item lemma-abelian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-stalk-module-differentials: sources/stacks-modules/pinned-original-source.tex, source lines 5197–5197. Navigation pointer only: the counted text refers to the Stacks item lemma-stalk-module-differentials, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference algebra-lemma-NL-homotopy: sources/stacks-modules/pinned-original-source.tex, source lines 5974–5974. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-NL-homotopy, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-section-netherlander: sources/stacks-modules/pinned-original-source.tex, source lines 5827–5827. Navigation pointer only: the counted text refers to the Stacks item algebra-section-netherlander, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *In the situation above there is a canonical isomorphism* $NL(\alpha) = NL_{\mathcal{B}/\mathcal{A}}$ *in* $D(\mathcal{B})$.

We can choose a different presentation of $\mathcal{B}$ as a quotient of a polynomial algebra over $\mathcal{A}$ and still obtain the same object of $D(\mathcal{B})$. To explain this, suppose that $\mathcal{E}$ is a sheaves of sets on X and $\alpha : \mathcal{E} \to \mathcal{B}$ a map of sheaves of sets. Then we obtain an $\mathcal{A}$-algebra homomorphism $\mathcal{A}[\mathcal{E}] \to \mathcal{B}$. If this map is surjective, i.e., if $\alpha(\mathcal{E})$ generates $\mathcal{B}$ as an $\mathcal{A}$-algebra, then we set

$$NL(\alpha) = \left(\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega_{\mathcal{A}[\mathcal{E}]/\mathcal{A}} \otimes_{\mathcal{A}[\mathcal{E}]} \mathcal{B}\right)$$

where $\mathcal{J} \subset \mathcal{A}[\mathcal{E}]$ is the kernel of the surjection $\mathcal{A}[\mathcal{E}] \to \mathcal{B}$.

Proof. Observe that $NL_{\mathcal{B}/\mathcal{A}} = NL(\mathrm{id}_{\mathcal{B}})$. Thus it suffices to show that given two maps $\alpha_i : \mathcal{E}_i \to \mathcal{B}$ as above, there is a canonical quasi-isomorphism $NL(\alpha_1) = NL(\alpha_2)$ in $D(\mathcal{B})$. To see this set $\mathcal{E} = \mathcal{E}_1 \amalg \mathcal{E}_2$ and $\alpha = \alpha_1 \amalg \alpha_2 : \mathcal{E} \to \mathcal{B}$. Set $\mathcal{J}_i = \mathrm{Ker}(\mathcal{A}[\mathcal{E}_i] \to \mathcal{B})$ and $\mathcal{J} = \mathrm{Ker}(\mathcal{A}[\mathcal{E}] \to \mathcal{B})$. We obtain maps $\mathcal{A}[\mathcal{E}_i] \to \mathcal{A}[\mathcal{E}]$ which send $\mathcal{J}_i$ into $\mathcal{J}$. Thus we obtain canonical maps of complexes

$$NL(\alpha_i) \longrightarrow NL(\alpha)$$

and it suffices to show these maps are quasi-isomorphism. To see this it suffices to check on stalks (Lemma lemma-abelian). If $x \in X$ then the stalk of $NL(\alpha)$ is the complex $NL(\alpha_x)$ of Algebra, Section algebra-section-netherlander associated to the presentation $\mathcal{A}_x[\mathcal{E}_x] \to \mathcal{B}_x$ coming from the map $\alpha_x : \mathcal{E}_x \to \mathcal{B}_x$. (Some details omitted; use Lemma lemma-stalk-module-differentials to see compatibility of forming differentials and taking stalks.) We conclude the result holds by Algebra, Lemma algebra-lemma-NL-homotopy. □